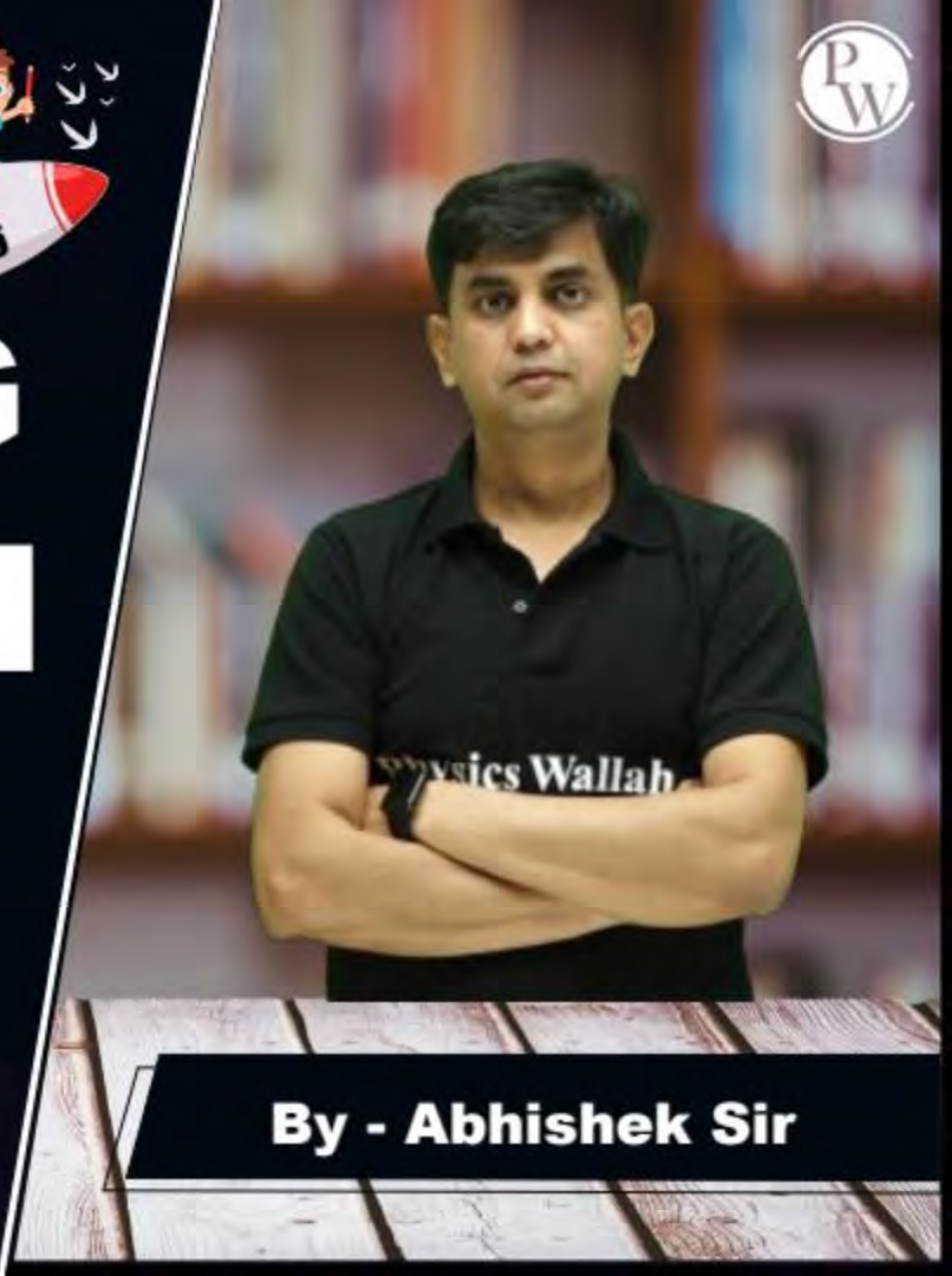
CS & IT ENGINEERING

Computer Network - 1

Transport Layer

Lecture No. – 03



By - Abhishek Sir

Recap of Previous Lecture

Topic UDP

Topic TCP

Topics to be Covered

Topic TCP Header

Topic TCP Connections

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : TCP

=> Connection-oriented :

-> Need to establish (logical) connection between TCP sender and receiver

- 3-way handshaking for connection establishment
- 4-way handshaking for connection close

-> In order delivery of messages to the communicating process

Topic : TCP

=> Reliable :

-> Recover lost messages, via retransmission

=> Complex and Slow :

-> Need to maintain connection state at TCP sender and receiver

-> Provide flow and congestion control

Topic : TCP

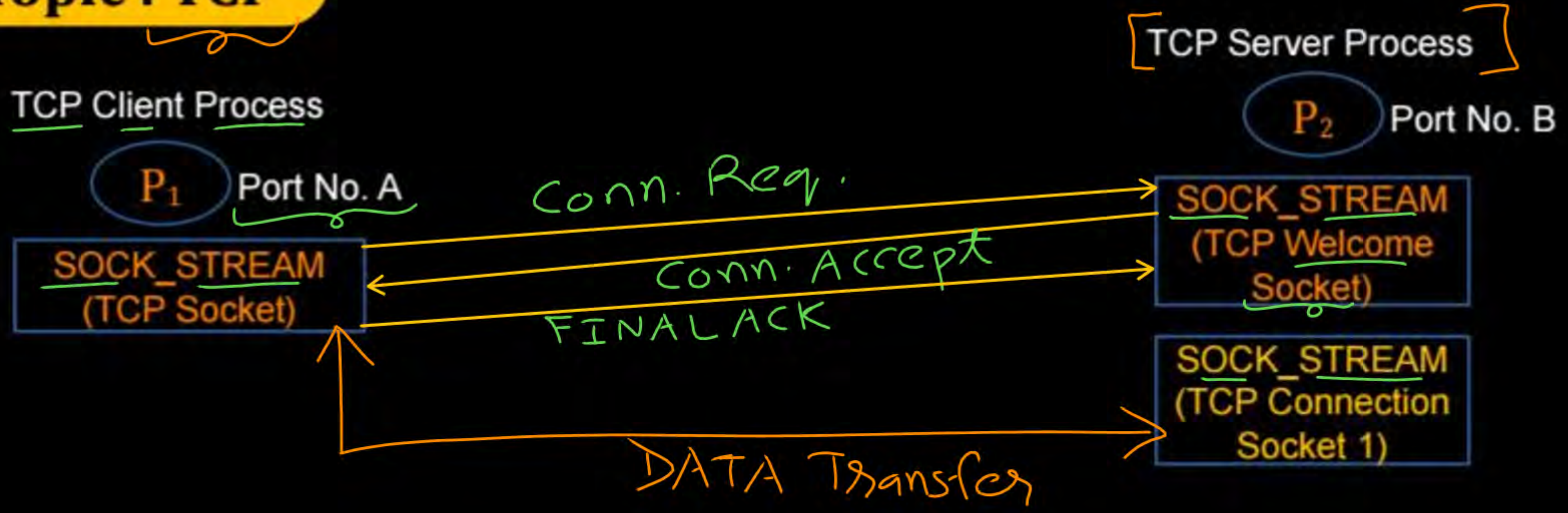
-> TCP : Transmission Control Protocol

-> Full-duplex and point-to-point logical connection

-> Provide Flow, Error and Congestion Control

-> Prefer for long communication

Topic : TCP


TCP Server Process
TCP Client Process
P1
Port No. A
P2
Port No. B
Conn. Req.
Conn. Accept
FINAL ACK
SOCK_STREAM
(TCP Socket)
SOCK_STREAM
(TCP Welcome Socket)
SOCK_STREAM
(TCP Connection Socket 1)
DATA Transfer

Topic : TCP

TCP Server Process

P_2 Port No. B

SOCK_STREAM (TCP Welcome Socket)

SOCK_STREAM (TCP Connection Socket 1)

SOCK_STREAM (TCP Connection Socket 3)

TCP Client Process

P_1 Port No. A

SOCK_STREAM (TCP Socket)

TCP Client Process

P_3 Port No. C

SOCK_STREAM (TCP Socket)

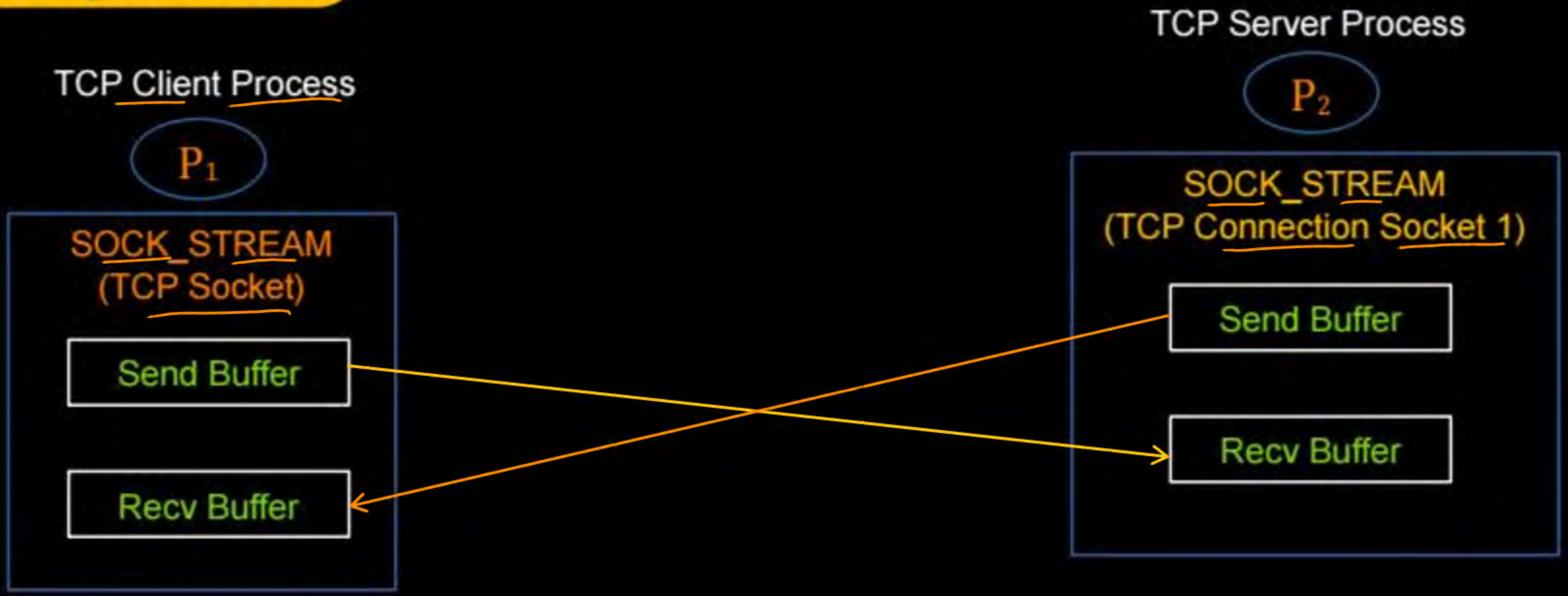
DATA Transfer

Topic : Connection-Oriented Demultiplexing

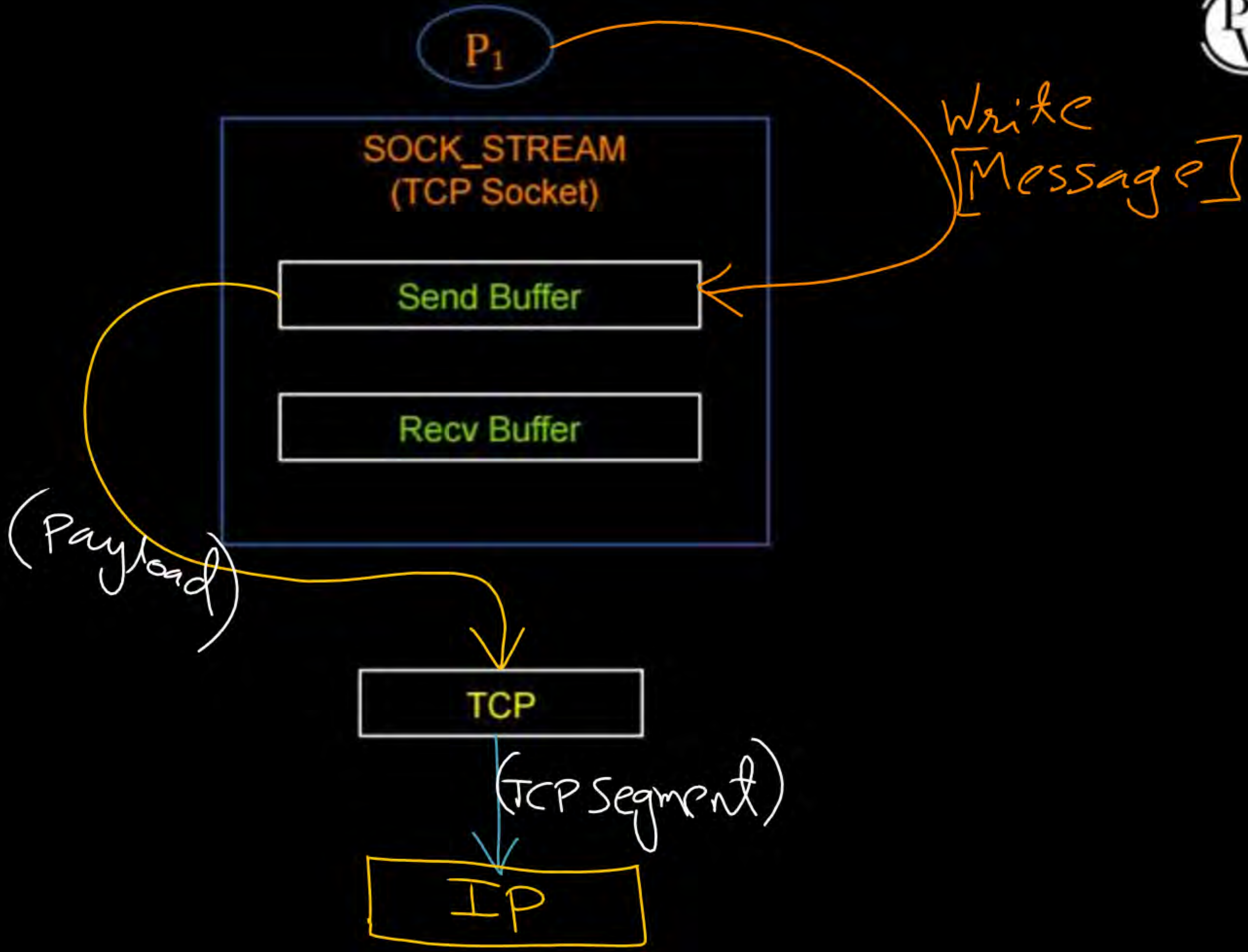
-> TCP sockets identified by 4 tuples :

Destination IP address	and	Destination Port Number
Source IP address	and	Source Port Number

Topic : TCP

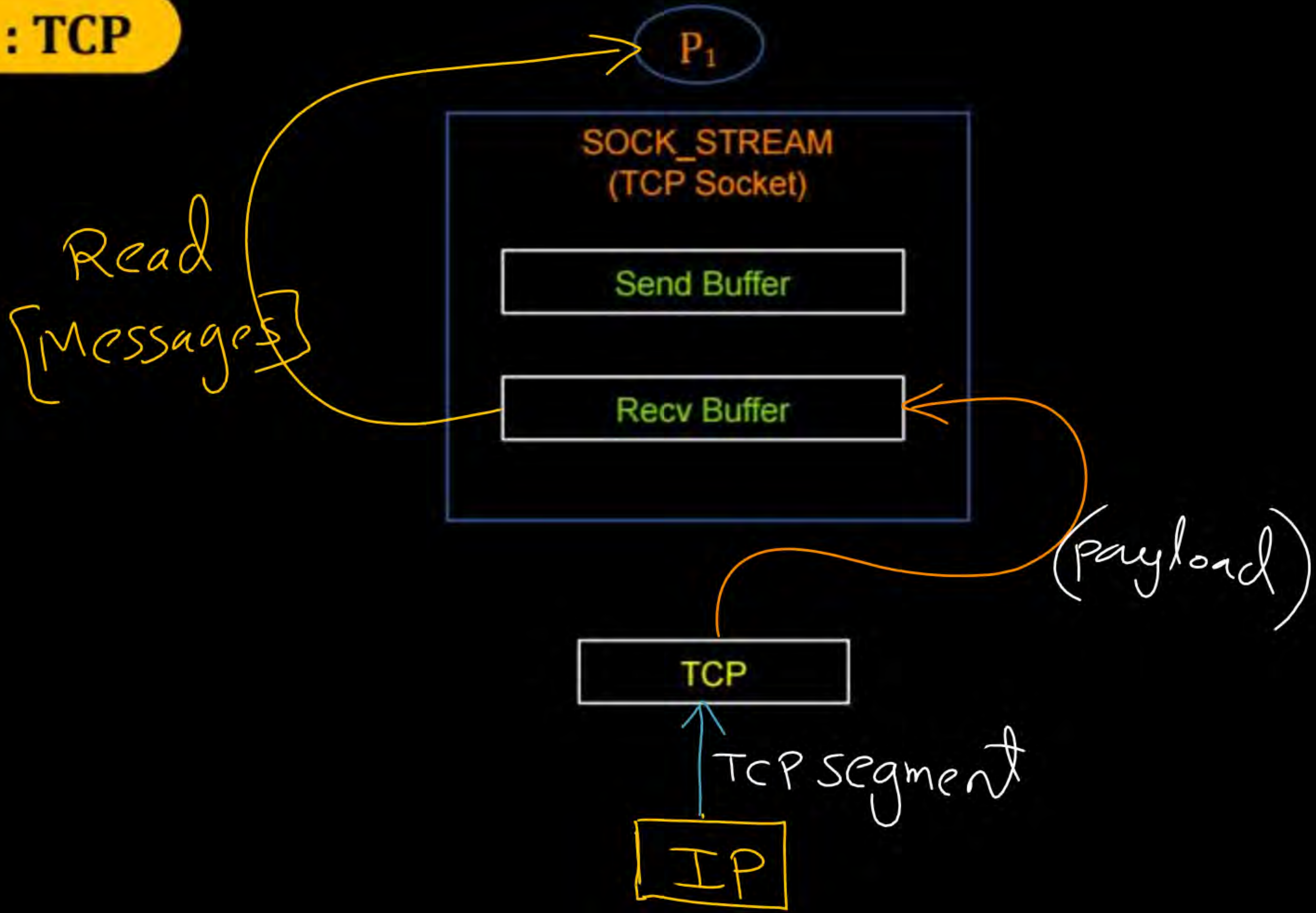

TCP Client Process
P_1
SOCK_STREAM
(TCP Socket)
Send Buffer
Recv Buffer
TCP Server Process
P_2
SOCK_STREAM
(TCP Connection Socket 1)
Send Buffer
Recv Buffer

Topic : TCP

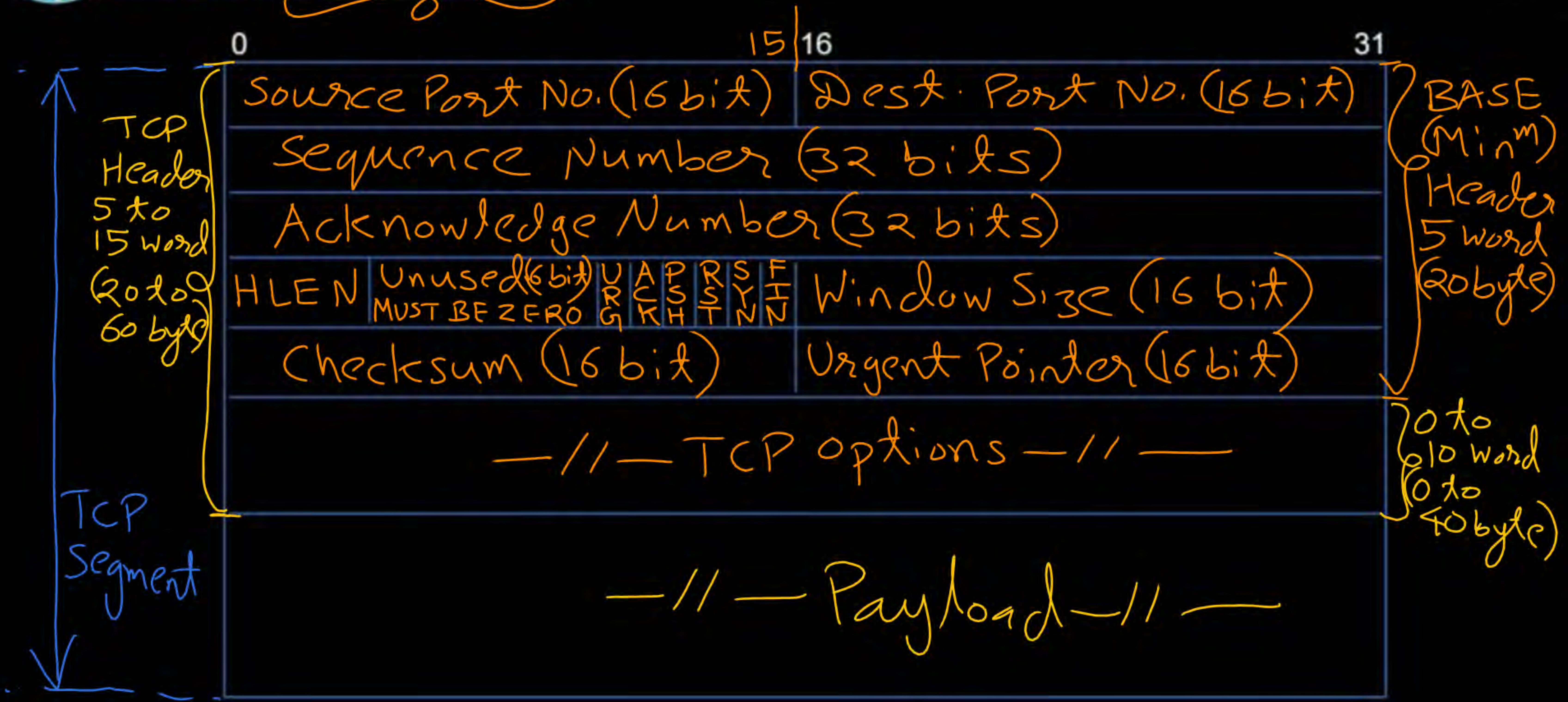


P1
Write [Message]
SOCK_STREAM
(TCP Socket)
Send Buffer
Recv Buffer
(Payload)
TCP
(TCP Segment)
IP

Topic : TCP


P1
SOCK_STREAM
(TCP Socket)
Send Buffer
Recv Buffer
Read
[Messages]
(payload)
TCP
TCP segment
IP

Topic : TCP Header



Topic : TCP Header

=> Source port number : 16-bits

=> Destination port number : 16-bits

=> Sequence Number : 32-bits

=> Acknowledgment Number : 32-bits

→ TCP support piggybacking of ACK.

Topic : Header Length

→ Header Length [HLEN]

→ HLEN field is 4 bits long

→ Size of header in words
[Word of 4 bytes]

Topic : Header Length

→ $\underbrace{\text{Minimum Header Size}}$ = 5 Words (20 Bytes)

$\boxed{5 <= \text{HLEN} <= 15}$

→ $\underbrace{\text{Maximum Header Size}}$ = 15 Words (60 Bytes)

Topic : TCP Flag Bits

=> URG	:	Urgent
=> ACK	:	Acknowledgement
=> PSH	:	Push
=> RST	:	Reset
=> SYN	:	Synchronize
=> FIN	:	Finish

Topic : Checksum

–> Checksum : 16-bits

–> 16-bit one's complement of the one's complement sum of all 16-bit words in TCP segment including pseudo header

–> Error detection in TCP is compulsory [Unlike UDP]

Topic : IPv4 Pseudo-Header $\Rightarrow$ 12 bytes

0	16	31
Source IPv4 Address (32 bits)		
Destination IPv4 Address (32 bits)		
Reserved	Protocol = 6	TCP Segment Length (16 bits)

Topic : TCP Header

0			16	31
Source Port Number (16 bit)			Destination Port Number (16 bit)	BASE Header
Sequence Number (32 - bits)				
Acknowledgment Number (32 - bits)				
HLEN	Unused Must be zero	6 Flag bits	Window Size (16 bit)	
Checksum (16 bit)			Urgent Pointer (16 bit)	
Optional Header				
Payload				

Topic : TCP Header

IPv4 Pseudo Header (12 bytes)

TCP Header (20 - 60 bytes)
[Including Optional Header]

Payload

Topic : TCP Operation

Three phases of TCP operation :

1. Connection establishment

   → 3-way handshake process that establishes a connection

2. Data transfer

3. Connection termination (CS-17)

   → 4-way handshake process that closes the connection

CN-2

Topic : TCP Connection Establishment

→ Connection establishment between TCP client and TCP server

→ 3-way handshake process

→ Always TCP client initiate the connection request to TCP server

2 mins Summary

Topic TCP Header

Topic ~~TCP Connections~~

THANK - YOU

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